

Modeling The Cognitive Process of Evidence-Accumulation for Hierarchical Decision-making
in a 4-choice Task Paradigm

Honors Thesis Submitted by
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in partial fulfillment of the
Sc.B. in Computational Cognitive Neuroscience
Brown University

Apr 18, 2022

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Abstract

This thesis aims to investigate the nature of hierarchical decision-making via generative cognitive process models. Focus is placed on three kinds of such process models, all in the domain of sequential sampling models (SSMs), a class of models that produce choice and reaction time behavior via diffusion to a bound mechanism. We consider a model that represents fully parallel processing, a model that represents fully sequential processing, and finally, a resource-tradeoff model which represents a mixture of parallel and sequential processing. The models were fit to behavioral data collected on a perceptual hierarchical decision-making task. To facilitate inference on these SSMs, we exploited recent advances in likelihood-free inference algorithms. Specifically, we made use of likelihood approximation networks (LANs). We demonstrate that the models capture many aspects of the observed behavior and thoroughly discuss where they still fall short of a complete account of the data. The resource-tradeoff model, which produces the best fit to the data, has some shortcomings concerning parameter identifiability, which we illustrate via parameter recovery studies. Lastly, we consider variations of all SSMs that include a collapse in the decision threshold over time, which incorporates an “urgency” signal into the generative models. We conclude that such collapsing bounds did not significantly improve our model fits.

Chapter 1: Introduction

Background:

Real-world decisions are often dependent upon prior decisions or observations. For example, let us say you are trying to pick a T-shirt to wear today, and it is cold outside, so you will only wear a long-sleeve T-shirt. On one hand, you can consider all of the T-shirts in your wardrobe, comparing all of them to see which one you like and is also long-sleeved. On the other hand, you can first identify all of the long-sleeve T-shirts, then see which of those shirts you like the best. In the latter case, you are employing a hierarchical decision-making strategy to avoid judging how much you like shirts that you will not wear because it is too cold. While many perceptual decision-making studies look at behaviors in a simple, independent task, less is known about how people make perceptual decisions when the task they are completing is contingent upon a decision in a different task. People may employ certain hierarchical evidence accumulation strategies to efficiently make such decisions, and the objective is to identify the most likely strategies involved in perceptual hierarchical decision-making.

Kim et al. (2017) designed a task to address if and how certain evidence accumulation strategies are utilized for hierarchical decision-making. The basic idea is the following (the task will be described in detail in the next section). Subjects are presented with a choice screen that displays information about three tasks that may be solved simultaneously (see Figure 1). However, one of these tasks is a priori designated as the “high dimension” task, and dependent on the respective solution to this task, the corresponding “low dimension” task is picked from the remaining two (see Figure 2). To prevent the paradigm from artificially encouraging sequential processing, the presented stimulus has the features required to answer all the tasks right from the beginning, and the participants have four possible choices, where a single response represents the

answers to the high dimension and low dimension tasks at once, as shown in Figure 2. The previous analysis from Kim et. al. (2017) attempted to determine if the behavioral data better aligned with sequentially processing each task (i.e. complete the high dimension task, then complete the low dimension task) or processing in parallel (i.e. accumulating evidence for all tasks at the same time).

The researchers attempted to determine the better model through behavioral analysis, comparing their observations to predictions of sequential and parallel sampling models. Fitting and comparing models was infeasible at the time due to the computational demands involved in performing inference for non-standard sequential sampling models. Without access to analytical likelihood functions for these non-standard sequential sampling methods, only likelihood-free inference methods, which require only a simulator, can be applied. These techniques, however, traditionally come at a great computational cost (Cranmer et al., 2020). However, the development of likelihood-free inference algorithms is an active area of research with many recent advances, which we can use to aid us (Greenburg et al., 2019; Gutmann et al., 2017; Papamakarios et al., 2016; Radev et al., 2020).

The present study utilizes the HDDM python package (Wieki et al., 2013) to fit and compare sequential sampling models of hierarchical decision-making. HDDM is widely used for purposes of hierarchical bayesian modeling of drift-diffusion models (basic forms of SSMs) (Ratcliff & McKoon, 2008; Ratcliff et al., 2016). However, hierarchical decision-making tasks have yet to be modeled using HDDM. To enable this we exploit a recent extension to HDDM, which incorporates the ability to use likelihood approximation networks to perform Bayesian inference on custom models (Fengler et al., 2021). By training neural networks on kernel density estimates built from random samples of our SSMs, these neural networks can act as approximate

likelihood functions for models that otherwise lack such likelihood functions in analytical form and are too expensive to use traditional likelihood-free methods for parameter recovery.

The prior analysis suggests sequential processing coupled with urgency would capture much of the observed behavior. While there is evidence for urgency via collapsing boundaries in perceptual decision-making, the support for them is mixed, and they may not improve the model (Hawkins et al., 2015). The current analysis does not see collapsing boundaries improving the model fits, but this may be due to higher model flexibility than expected in Kim et al., (2017).

Task:

The hierarchical decision-making task consists of a collection of lines being presented on a computer screen (see Figure 1). These lines have three features: orientation, color, and movement direction. Each of these features only has two possible states, allowing for binary decisions as far as single features are concerned. As seen in Figure 2, orientation (angle) has the options of a forward slash or a backward slash, color has the options of yellow or pink, and movement direction has the options of left or right. One of these features would constitute the high dimension task while the other two would be the low and irrelevant dimensions. The high dimension task was fixed for each participant, while the low and irrelevant dimensions would flip according to the correct response for the high dimension task. The low dimension task is what the participant responds to while the irrelevant dimension task is the task for which the participant does not provide a response. Rather than respond to each dimension separately, the participants only give one response out of four possible choices, which represents their answer to both the low and high dimensions. The choices and their respective reaction times were collected for each trial and form the basis of the analysis in this thesis.

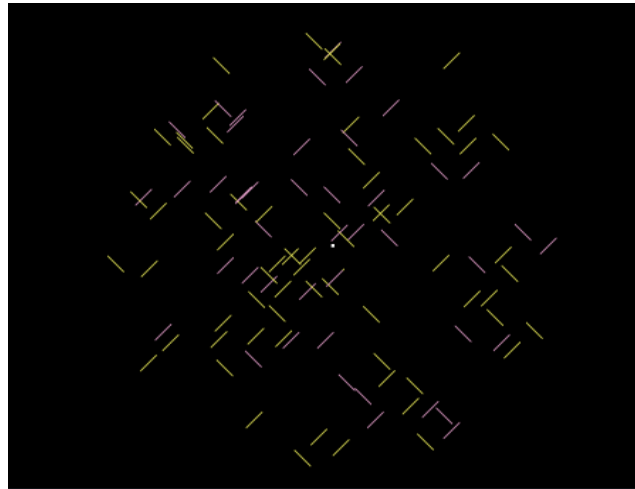


Figure 1: From Kim et al. (2017). Example of stimuli presented to participants on a computer screen. The lines on the screen can either be forward or backward slashes, be yellow or pink, and move either to the right or left. For each of these features, one of the two states is greater in presence.

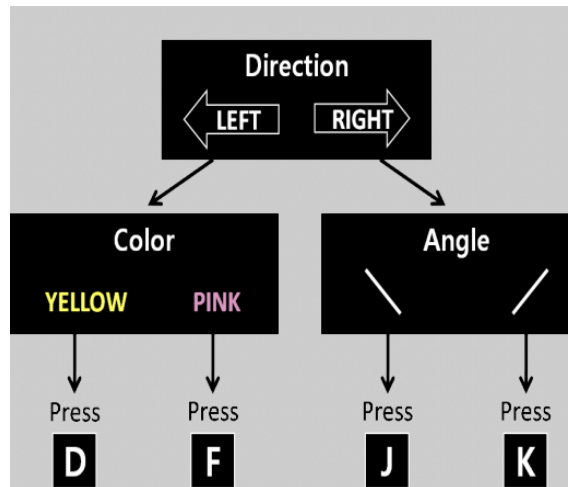


Figure 2: From Kim et al. (2017). Example diagram of hierarchical task design. In this example, the direction the lines move determines which low dimension task to complete and which set of two buttons are considered. The response

to the low dimension task then determines which button to press. The pressed button, therefore, represents the response to both the direction and the appropriate low dimension task in this example.

Each of the three tasks varies between two coherence conditions. In each of the tasks, only a certain percentage of the lines have the state for the correct answer. The participant must respond to the dominant attribute for each task to be correct. The coherence, or percentage of lines with the same attribute, varies between the conditions of high and low coherence. The high coherence condition has a greater majority of the lines sharing the same attribute for a given task, while low coherence has a smaller percentage in the majority. These coherence conditions, therefore, change the task difficulty, allowing for comparative analyses between an easier and a harder version of the same task. For example, the color condition in this experiment has 70% of the lines being the dominant color with 30% being the other color in the high coherence condition while, in the low coherence condition, 58% are the dominant color with 42% being the other color.

There are three types of errors to be made in this task: (1) correct on the high dimension, incorrect on the low dimension (low dimension error); (2) incorrect on the high dimension, correct on the low dimension (high dimension error); (3) incorrect on both high and low dimensions. The current analysis focuses on the first two errors. The third error type is rare and harder to interpret. Importantly, high dimension errors do not correctly complete the intended low dimension but rather the intended irrelevant dimension.

In a given trial, the high, low, and irrelevant dimensions each have independent coherence levels. When analyzing the data, only the coherence of the completed tasks is viewed because the regression analysis by Kim et al. (2017) indicated no meaningful effect of the

irrelevant dimension's coherence on reaction times (see Figure 7), though the effect of the irrelevant dimension's coherence was not thoroughly examined. This creates four coherence categories: High coherence in the high dimension, high coherence in the low dimension (HH); low coherence in the high dimension, high coherence in the low dimension (LH); high coherence in the high dimension, low coherence in the low dimension (HL); low coherence in the high dimension, low coherence in the low dimension (LL). These four coherence conditions separate the data for analysis and are important for viewing the effects of hierarchical processing.

One potential confound in this task is differences in task difficulty. Ideally, analysis of data from this experiment would not include task effects, but there is no guarantee that these tasks are equivalent. Such a confound could affect the importance of coherence levels and skew reaction times. While the behavioral analysis does not examine differences between the tasks, the models presented can, to some extent, vary their behavior for each task. (the different tasks can be seen in Figure 2).

Goal:

The goal of this study is to explain and reproduce the observed behavior through computational cognitive process models of hierarchical evidence accumulation strategies. To this end, three strategies are considered via variations of SSMs: parallel processing, sequential processing, and trade-off processing that can be both somewhat sequential and parallel. Multiple variations of the models are considered. The models can vary by the type of decision boundary, the coherence level, and the task. Because of prior analysis from Kim et al., (2017) supporting collapsing bounds in modeling performance for this experiment, the three types of boundaries

considered are constant bounds, linearly decreasing bounds, and bounds shaped as a Weibull survival function.

Chapter 2: Behavioral Analysis

Previous Analysis:

Before the current work presented here, Ji Sun Kim, under the supervision of Dr. Matthew Nassar and Dr. Michael Frank, conducted behavioral analysis and EEG regression (the EEG analysis will not be addressed in this thesis). The initial conclusions regarding the observed behavior are as follows: (1) people adopt hierarchical evidence accumulation strategies during complex decisions to avoid unnecessary evidence accumulation, (2) subjects prioritize high dimensional evidence in the hierarchical perceptual decision-making task, (3) reaction time and error rates are consistent with sequential processing with a collapsing bound (Kim et al., 2017). By fitting models of hierarchical evidence accumulation and decision-making, these conclusions are further investigated.

The first and second conclusions are supported by the overall error rates seen in the data. In all coherence conditions and overall across the five computed quantiles, there are significantly more low dimension errors than high dimension errors (see Figures 3 and 8). In the overall error rate data, the difference between first and last quantiles of low dimension errors is much greater than the high dimension errors, as seen in Figure 3. The a priori expectation was that subjects would only show meaningful differences in behavior between low and high dimension errors if they were employing some serial evidence accumulation strategy. These differences also indicate favoring high dimension evidence as more time elapses.

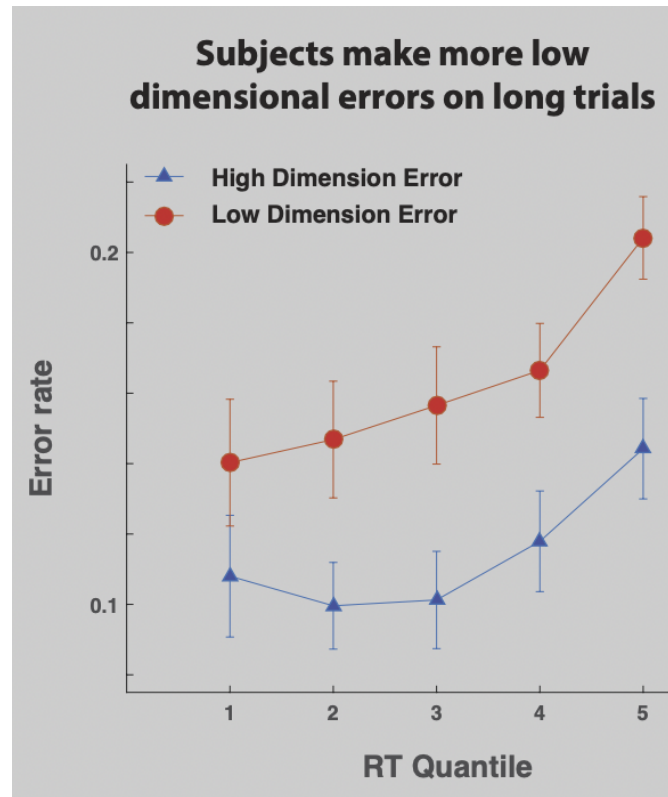


Figure 3: Figure from Kim et al. (2017). Plot of the proportion of overall high dimension and low dimension error rates across 5 reaction quantiles. The reaction time quantiles and number of errors for each quantile bin are calculated for each participant and then aggregated within the appropriate quantile.

The third conclusion is supported by predictions of parallel processing and increasing error rate in later reaction time quantiles. The previous analysis predicted parallel processing to show reaction times in the “LH” and “HL” coherence conditions closer to the reaction times in the “HH” condition than in the “LL” condition while sequential processing would show reaction times in “LH” and “HL” conditions to be equidistant from the “HH” and “LL” conditions (see Figures 4 and 5). This would be an interaction effect between the high dimension coherence and low dimension coherence. The subject data, however, aligns with the expectations of sequential

processing rather than parallel processing as it does not show an interaction effect (see Figures 5, 6, and 7). This expectation of parallel processing behavior assumes an equal strength of evidence accumulation in the high and low dimensions.

The previous analysis predicted sequential and parallel processing with constant boundaries would produce error rates that are equivalent in the high and low dimensions and across quantiles, but this too assumes equivalent evidence accumulation in the high and low dimensions. It was reasoned that the addition of an urgency signal (via collapsing bounds) could reproduce the observed phenomenon of increasing error rates across reaction time quantiles.

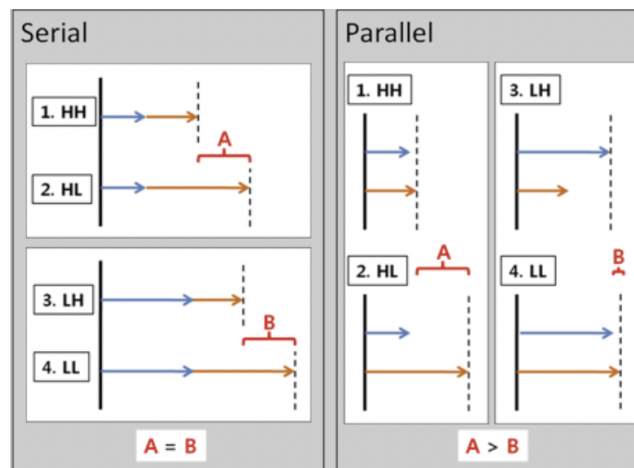


Figure 4: Figure from Kim et al. (2017). A priori expectation of reaction time differences between the parallel and sequential models of processing. The arrow length is the amount of time till a decision is made. “H” and “L” denote high and low coherence levels for the high and low dimension tasks. The blue arrow is the high dimension task, and the orange arrow is the low dimension task.

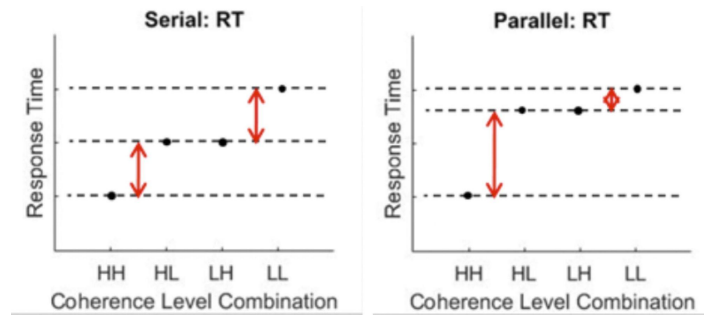


Figure 5: Figure from Kim et al. (2017). A priori expectation of reaction time differences between the parallel and sequential models of processing. This figure illustrates that parallel processing should have “HL” and “LH” coherence condition reaction times closer to the reaction times of the “LL” condition than the “HH” condition. Meanwhile, the serial model should have reaction times of the “HL” and “LH” coherence conditions be equidistant between the reaction times of the “HH” and “LL” coherence conditions.

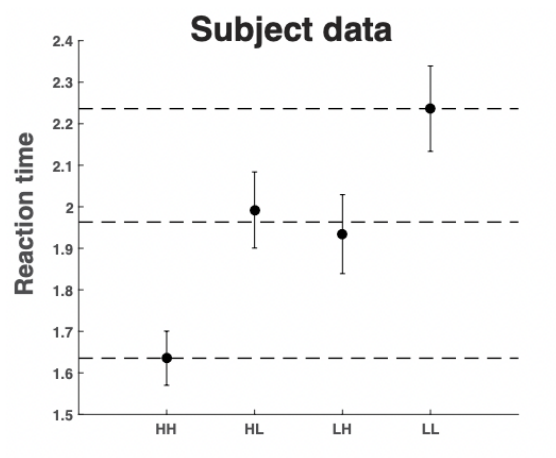


Figure 6: Figure from Kim et al. (2017). Mean reaction time by coherence condition of the aggregate subject data.

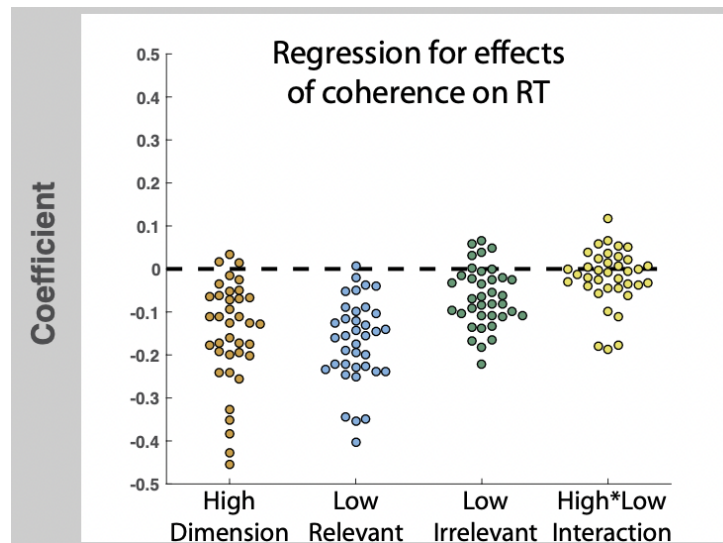


Figure 7: From Kim et al. (2017). Plot of linear model slopes for high to low coherence condition reaction times by each task dimension.

Current Analysis:

The current behavioral analysis lightly builds upon the findings of the previous analysis, using the same types of plots with the addition of quantile probability plots. Quantile probability plots have been previously used to demonstrate behavior seen in reaction time and accuracy across conditions, specifically when examining drift-diffusion models (Ratcliff & Smith, 2010; Simen et al., 2009). The new analysis examines behavior primarily through quantile probability plots.

The purpose of quantile probability plots is to show the effect of coherence on reaction time distributions, teased apart by the specific accuracy of the decision taken in the high and low dimensions (incorrect on low coherence, correct on low coherence, incorrect on high coherence, correct on high coherence). The quantile probability plots in Figures 8, 16, 17, and 18 use 5 quantiles, which are represented as stacked dots. Starting from the bottom, the quantiles are the 0

quantile (minimum reaction time), the 0.2 quantile, the 0.4 quantile, the 0.6 quantile, and the 0.8 quantile. The quantile probability plots for the high dimension and the low dimension each display four reaction time spreads for each combination of the difficulty represented by the coherence level and the response type (correct or incorrect). The location of a reaction time spread on the x-axis represents the observed probability of that response within its given coherence level. The addition of the probabilities of correct and incorrect responses within a given coherence level will not add up to 1 because trials with errors on both dimensions are not plotted but were included in the probability calculation.

Certain data was removed from the analysis. All of subject 13's trials were excluded because this participant not only made the most errors out of any participant but also comprised the vast majority of errors in small reaction times. Subject 13 made 157 errors with a reaction time <0.5 seconds while the next highest number of errors with a reaction time <0.5 seconds in a subject was 17 (most subjects showed no errors in this time frame). Additionally, trials where a participant made an error in both the low and the high dimensions were not analyzed. This is because errors on both are rare in the dataset ($<4\%$) and is difficult to interpret.

In addition to previous findings, the current behavioral analysis identified an important behavior in the data: The proportion of high dimension errors is lower in conditions where the low dimension's coherence level is high compared to when the low dimension's coherence level is low. This effect suggests that the difficulty of the low dimension task affects performance not just on the low dimension but on the high dimension as well. While there was a prior expectation for high dimension difficulty to affect low dimension performance, as would be expected of collapsing bounds in a sequential processing model, the opposite was unexpected.

The quantile probability plots (see Figure 8) show that coherence has a relatively larger effect on reaction times for correct answers as compared to incorrect answers. The effects of coherence on reaction time and proportion of correct responses appear slightly stronger in the low dimension as well.

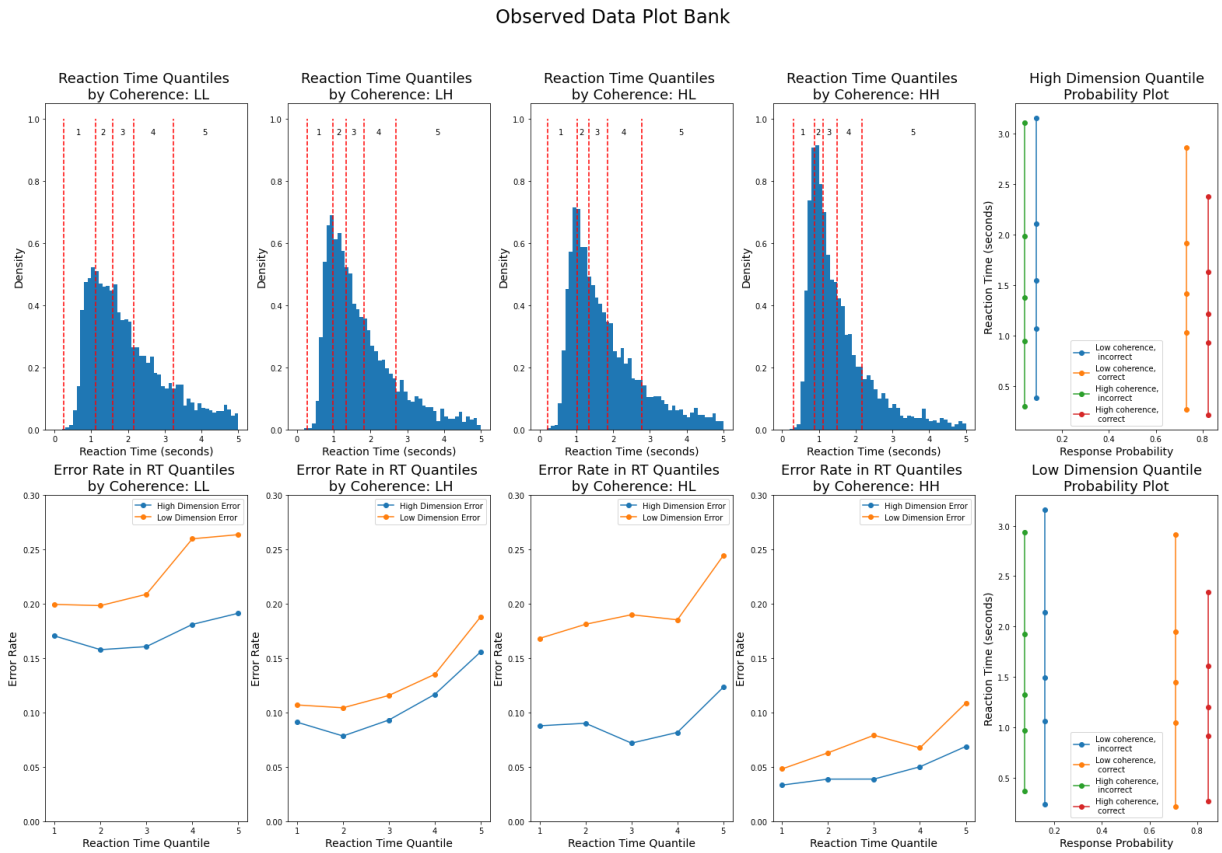


Figure 8: Plot bank of observed behavioral data. The reaction time histograms and errors by RT quantile are vertically aligned by their coherence condition. High coherence in the high dimension, high coherence in the low dimension (HH); low coherence in the high dimension, high coherence in the low dimension (LH); high coherence in the high dimension, low coherence in the low dimension (HL); low coherence in the high dimension, low coherence in the low dimension (LL). The red lines in the histogram plots mark the beginning of a new quantile, starting at the minimum reaction time (followed by the 0.2, 0.4, 0.6, and 0.8 quantiles). The error rate plots show the proportion of errors in each quantile, calculated by averaging the error rates in a quantile over all the subjects. The two quantile probability plots differ by the high dimension and low dimension. These plots have five dots in each line that represent the same quantiles as seen in the reaction time histograms.

Chapter 3: Models of Behavior

Overview:

To better understand and capture the behavior and strategy employed by the subjects to complete this task, models of certain strategies for evidence accumulation and decision-making were fit to the subject data and compared to both the real data and each other. This process aids research into how subjects are processing the task and identifying what may not be explained by some models and strategies. Because evidence accumulation is invisible, and only the end choice is accessible, these models allow insights into unobservable latent processes driving decision-making.

Using the HDDM (Hierarchical Drift-Diffusion Modeling) python package, three main models of behavior, and multiple variations of them, were fit to the subject data. These variations include flexible decision boundaries, which allow the boundary, which the accumulated evidence must cross to reach a decision, to change over time during the trial. The model parameters can also vary across conditions in the experiment. In this study, the model varied the drift rate parameter in each dimension by its matching task and coherence level and recovered all of the parameters, along with their variations, for each subject. In the model-fitting done so far, the most complex models fit new parameters for each subject and fit drift rates based on their relevant task and coherence level.

Models:

To best characterize the behavior seen in the subject data, three models of behavior were fit to the subject data: (1) Sequential, a model that only accumulates evidence for one task at a time; (2) Parallel, a model that accumulates evidence for all tasks; (3) Trade-off, a model that can

accumulate evidence sequentially and in parallel. All three models have the three drift parameters for the high, low, and irrelevant dimensions. The trade-off model has an additional dampen, or “d”, parameter that determines how parallel/sequential the model will accumulate evidence.

Two other important parameters are the threshold and non-decision time. The distance between the boundaries is determined by the threshold parameter. The non-decision time, or time not dedicated to evidence accumulation, is captured by the parameter non-decision time (e.g. motor execution). There are also starting-point bias parameters, but none of the current models are using bias parameters, so they all start in the center. Figure 9 below illustrates the effect of these parameters on the drift-diffusion model.

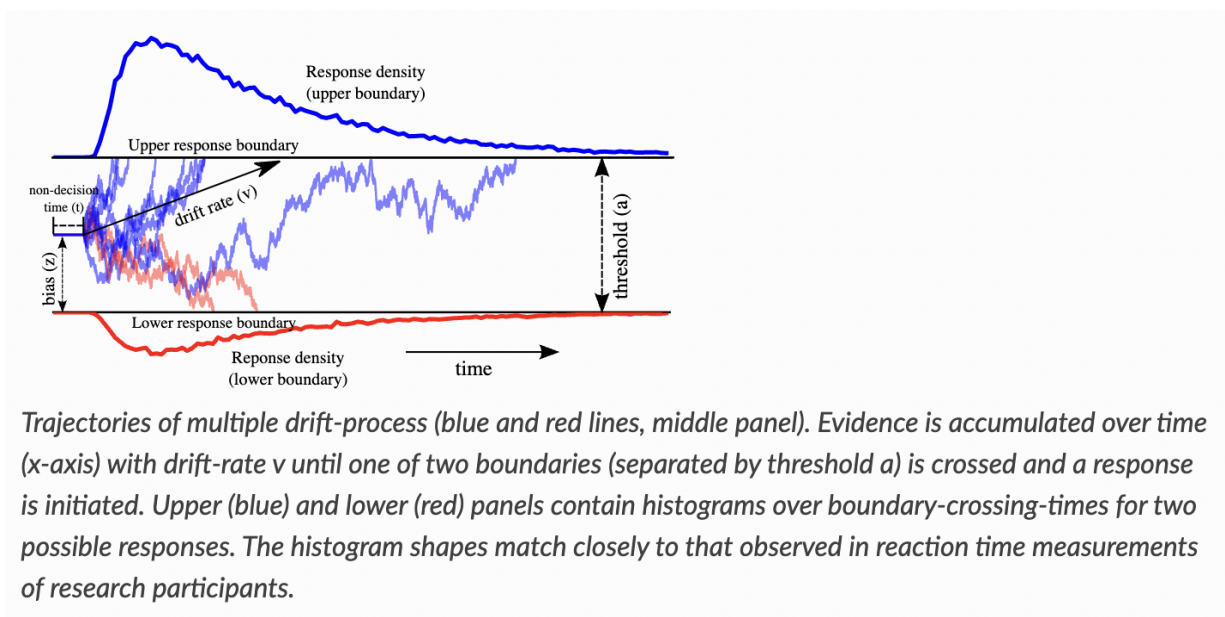


Figure 9: From Wiecki et al. (n.d.)

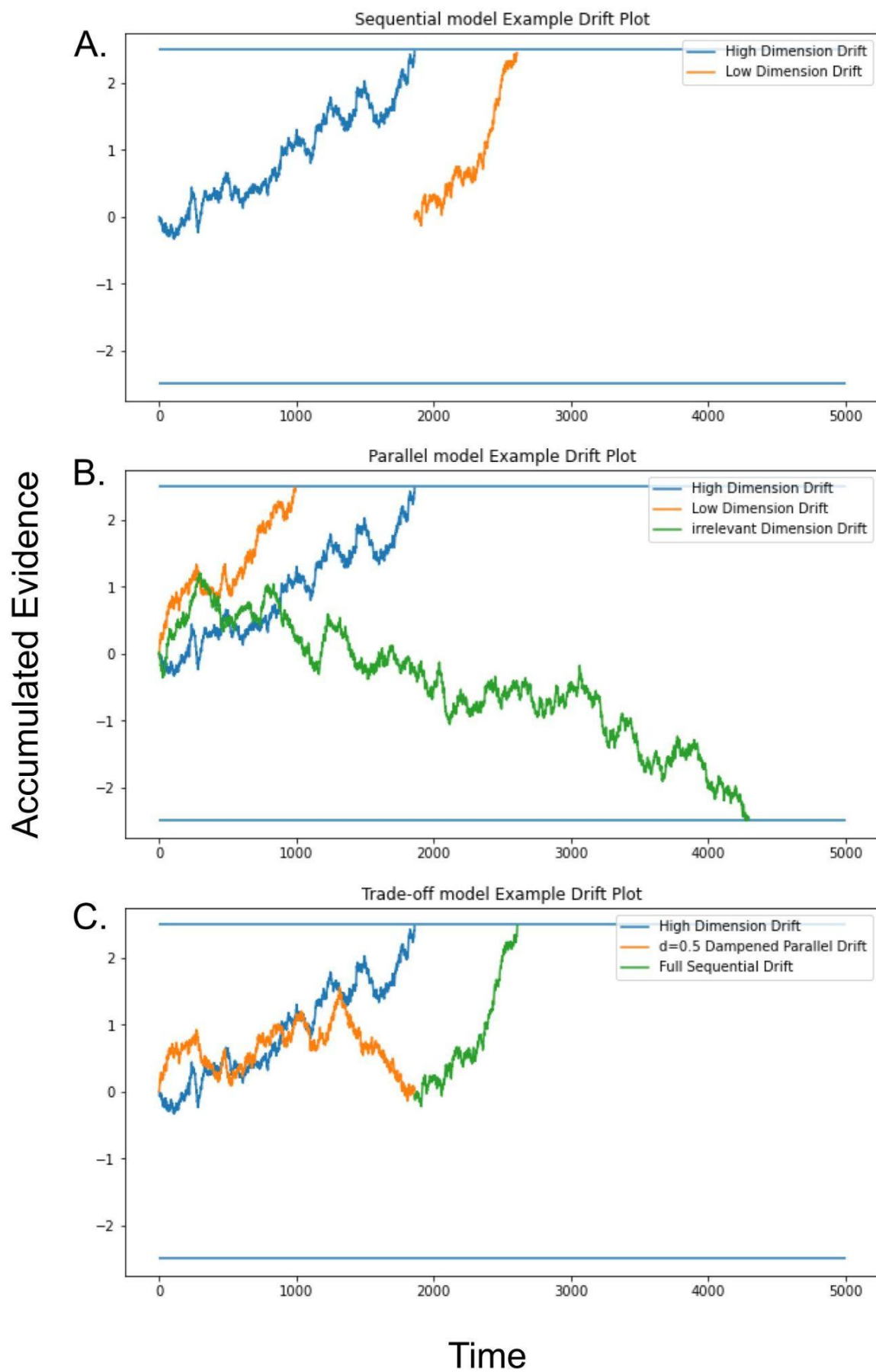


Figure 10: Plot of example evidence accumulation trajectories, illustrating the difference between the models. (A) In the sequential model, high dimension evidence accumulation reaches the decision boundary before the low dimension evidence accumulation begins. This avoids the accumulation of evidence for the irrelevant dimension. (B) In the parallel model, evidence accumulation for the high, low, and irrelevant dimensions begins at the same time. (C) In the trade-off model, evidence accumulation for the low dimension starts at the same time as the high dimension, but the influence of the low dimension drift rate is modulated by the dampen, or “d”, parameter and the distance of the high dimension accumulator from a decision boundary. The dampen parameter is a direct multiplier on the low dimension drift rate and can be any value from 0 to 1. Once the high dimension reaches a decision boundary, the low dimension accumulates evidence as done in the sequential model.

Models were also fit with either constant or flexible boundaries. Of the flexible boundaries, two have been implemented and fit, angle and Weibull. The “angle” flexible boundary takes one parameter, “theta”, which adjusts the slope of the boundary as a factor of time. For $\theta > 0$, the bounds ‘collapse’ (or shrink) over time. The other is the Weibull boundary. The Weibull boundary takes in shape parameter, “alpha”, and scale parameter, “beta”, and constructs a Weibull survival function to serve as the decision boundary. Figure 11 below shows an example of each boundary type and, to an extent, its effect on reaction time in the sequential model.

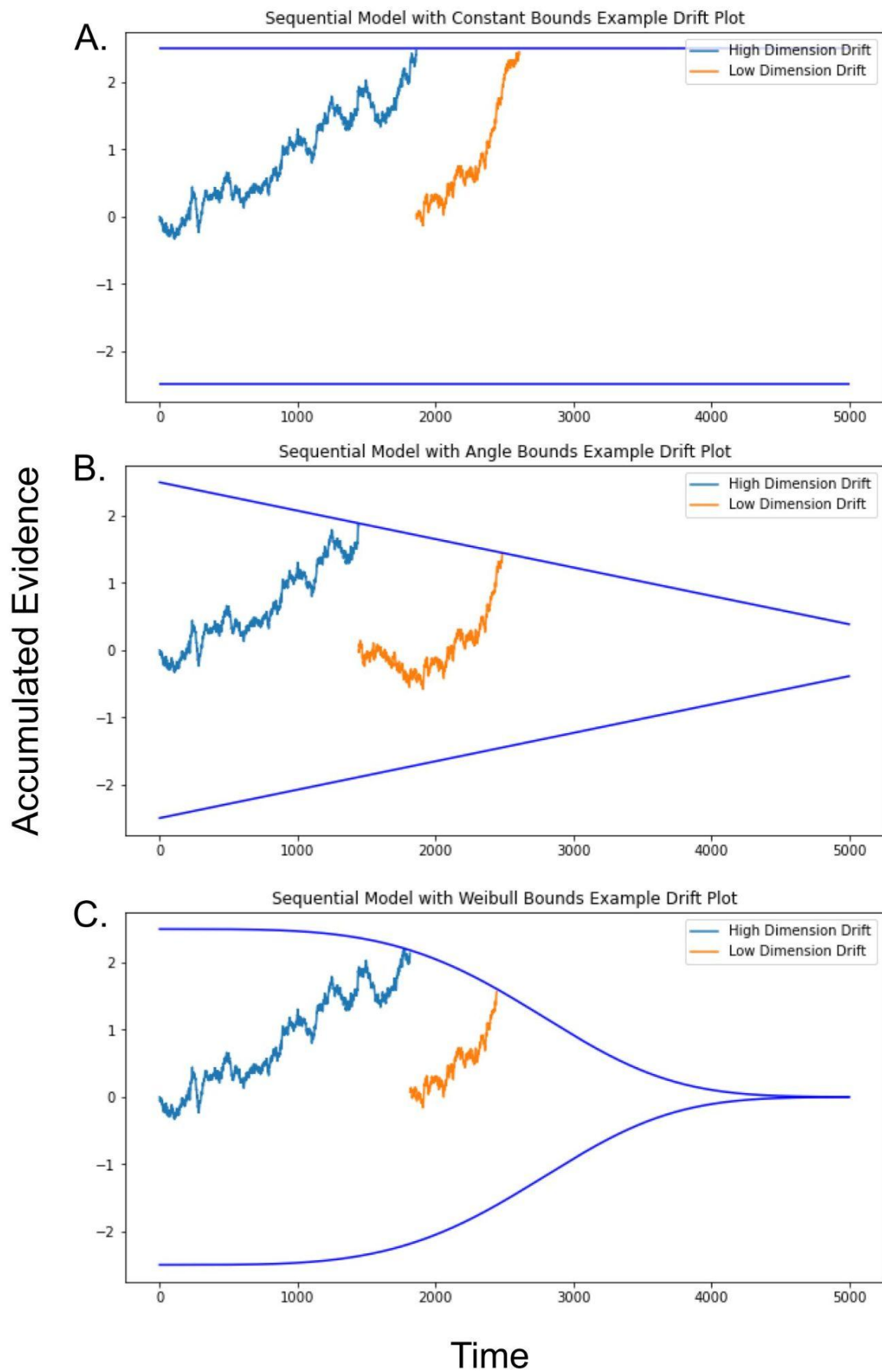


Figure 11: Plot of an example sequential evidence accumulation trajectory, illustrating the difference and effect of the three boundary types. (A) In the constant boundary type, the threshold stays the same throughout time and will, on average, produce the longest reaction times and have the highest accuracy out of the boundaries if all other parameters stay the same. (B) In the angle boundary type, both the low dimension and the high dimension tend to reach the boundary faster, but the effect is normally greater for the low dimension in a sequential model since it starts after the high dimension has finished. The boundary decreases linearly at an angle, θ . (C) In the Weibull boundary condition (which is a Weibull survival function), the low dimension tends to be affected significantly more than the high dimension in a sequential model due to its log-like shape change.

Sequential model:

The key behavior of the sequential model is reaching a decision in the high dimension before starting to accumulate evidence for the low dimension. The sequential model functions by adding both the high dimensional drift rate multiplied by a specified timestep and a random value sampled from a Gaussian distribution with a standard deviation that is the square root of the timestep to the current position of the accumulator. If the accumulator does not cross the boundary before 20 seconds, the decisions for both the low and high dimensions are sampled from a Bernoulli distribution. If the position crosses the negative boundary, then the low dimension drift rate is set as the parameter for the irrelevant dimension's drift rate. If the position crosses the positive boundary, then the low dimension drift rate is set as the parameter for the correct low dimension's drift rate. The low dimension decision is calculated the same way as the high dimension decision, and only the low dimension choice is sampled from a Bernoulli distribution if a high dimension choice has been made while the low dimension does not reach a boundary in time. See Figure 11.A for an example of sequential evidence accumulation compared to the other models. This model represents a hierarchical strategy as it avoids accumulating evidence for the irrelevant condition.

Parallel model:

The parallel model starts all three accumulators (high dimension, low dimension, and irrelevant dimension) at the same time. While the movement calculations are the same as the sequential model, the trial ends when the high dimension accumulator and accumulator that corresponds to the high dimension decision (either the low or irrelevant dimension accumulator) have both reached a decision boundary. The decision for each incomplete dimension is sampled from a Bernoulli distribution if time runs out. This model represents a non-hierarchical strategy as evidence is accumulated uniformly and simultaneously across all dimensions.

Trade-off model:

The trade-off model contains another parameter compared to the sequential and parallel models. The “d”, or dampen, parameter modulates the strength of evidence accumulation while a high dimension decision has not been made, with $d=1$ being maximally parallel and $d=0$ being maximally sequential, though the trade-off model cannot function exactly like either the parallel or sequential models. This is because the model still accumulates random evidence when fully dampened and it also modulates low dimension drift by the relative location of the high dimension particle when the model is not fully dampened. Figure 12 illustrates the effect of the dampen parameter.

The relative position of the high dimension accumulator is a value between 0 and 1 that represents the distance of the high dimension decision particle from a decision boundary, where 1 means it is at a bound and 0.5 means it is equidistant from the upper and lower bounds. Going below a value of 0.5 means that the high dimension accumulator position is closer to the bound it eventually does not choose. Whether a certain position is deemed as above or below 0.5 is

determined by the eventual high dimension choice. Whether the low dimension drift rate is for the irrelevant or correct low dimension is also determined once the high dimension choice is made. Effectively, the model goes back and applies the appropriate drift rate and relative positions to the low dimension drift after the high dimension choice is made. Once the high dimension reaches a decision boundary, low dimension evidence accumulates as it would without the dampen parameter and the high dimension accumulator's relative position, which is the same way as the sequential model. This can be seen in Figure 12, where the low dimension drift changes from orange to green once the high dimension reaches a boundary.

The equation below is how the effect of low dimension drift rate is calculated when accumulating low dimension evidence concurrently with the high dimension.

$$\text{low dim position} += (\text{low dim drift rate} * \text{high dim relative position} * \text{dampen}) * \Delta t$$

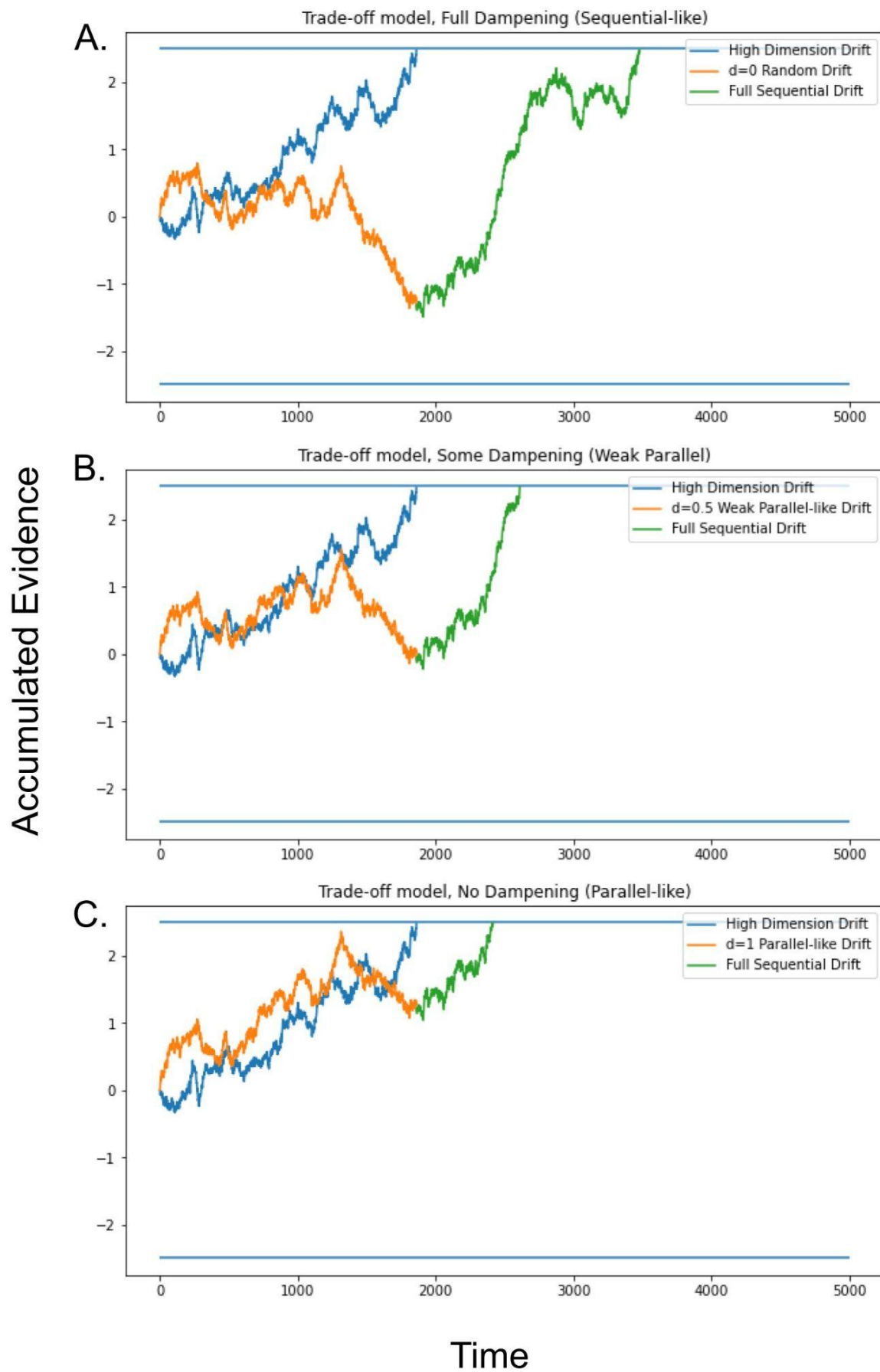


Figure 12: Plot of an example sequential evidence accumulation trajectory, illustrating the effect of the dampen, or “d”, parameter. (A) When the trade-off model’s low dimension concurrent evidence accumulation is fully dampened, the low dimension’s drift rate does not affect evidence accumulation, and the model walks randomly, driven only by Gaussian random sampling. Once the high dimension decision is made, the model accumulates low dimension evidence as though it were on the low dimension portion of the sequential model. (B) When the trade-off model’s low dimension concurrent evidence accumulation is half dampened, the low dimension drift plays a role such that the evidence accumulation is closer to the correct boundary than the random walk seen in A. (C) When there is no dampening, the model’s drift plays a larger role, as seen by the distance of the low dimension from the boundary before the high dimension has finished. Because the drift is also modulated by the relative position of the high dimension, however, the low dimension drift rate is not as influential as it would be in the parallel model.

Verification:

To ensure that this novel method of using surrogate neural networks for parameter recovery performs adequately, the models were fit to synthetic data. The specific verification pipeline is as follows: (1) A model is chosen (e.g. the sequential model); (2) Random parameters for that model are generated; (3) The random parameters are used to generate data of the same format as the subject data (i.e. synthetic data contains the same number of subjects and trials per subject as the subject data); (4) The model is fit to synthetic data using Markov chain monte Carlo (MCMC) with 2000 samples after a burn-in period of 1000 samples. This was done 120 times for each model, and the recovered parameters are plotted against the original parameters (Figures 13, 14, and 15).

The scatter plots seen in Figures 12, 14, and 15 show successful parameter recovery for all parameters except for the trade-off model’s dampen parameter. The plots also show that there is a tendency for the fit to overestimate the drift parameters as the original parameters approach the upper bound as well as a tendency to underestimate them closer to the lower bound. Having more parameters, as seen from the difference between the sequential/parallel models and the trade-off model, seems to have relatively worsened the recovery of the drift parameters. The

threshold and non-decision time parameters appear well recovered regardless of the number of parameters.

Trade-off Model Synthetic Parameter Recovery

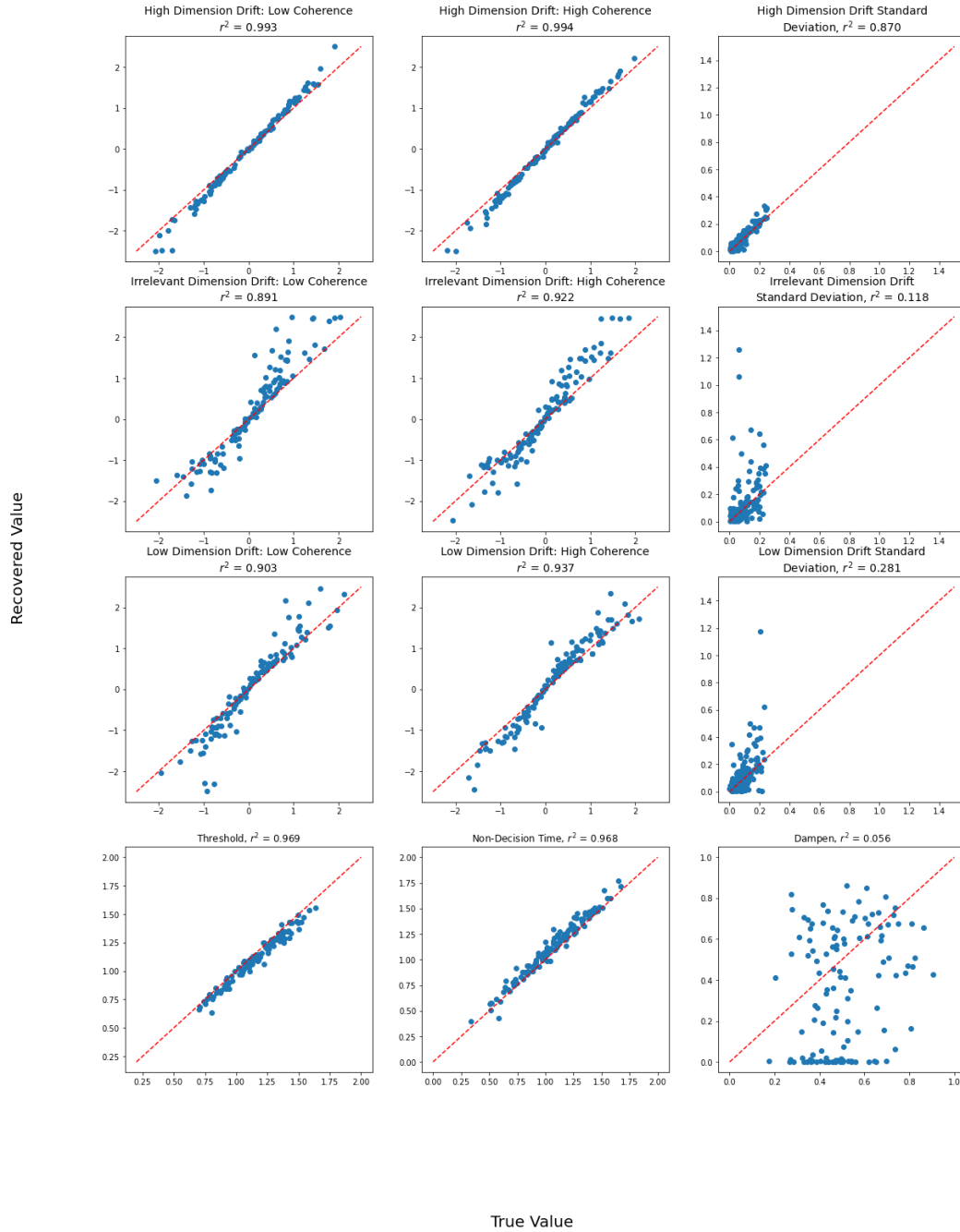


Figure 13: Scatter plots of each parameter's recovery across all 120 synthetic trials for the

trade-off model. The red line is the line of perfect recovery. Drift parameters are varied by the dimension's coherence level.

Sequential Model Synthetic Parameter Recovery

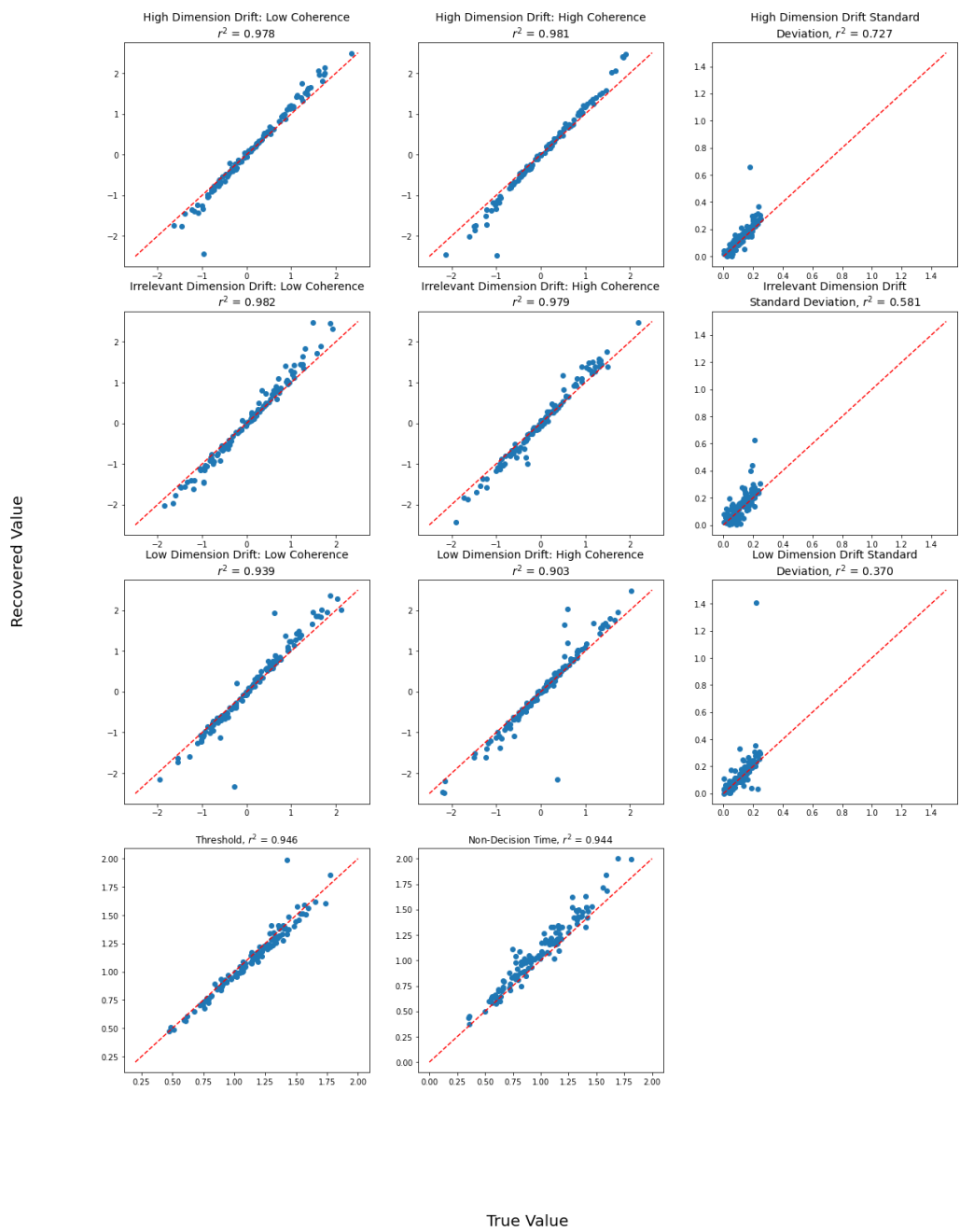


Figure 14: Scatter plots of each parameter's recovery across all 120 synthetic trials for the sequential model. The red line is the line of perfect recovery.

Parallel Model Synthetic Parameter Recovery

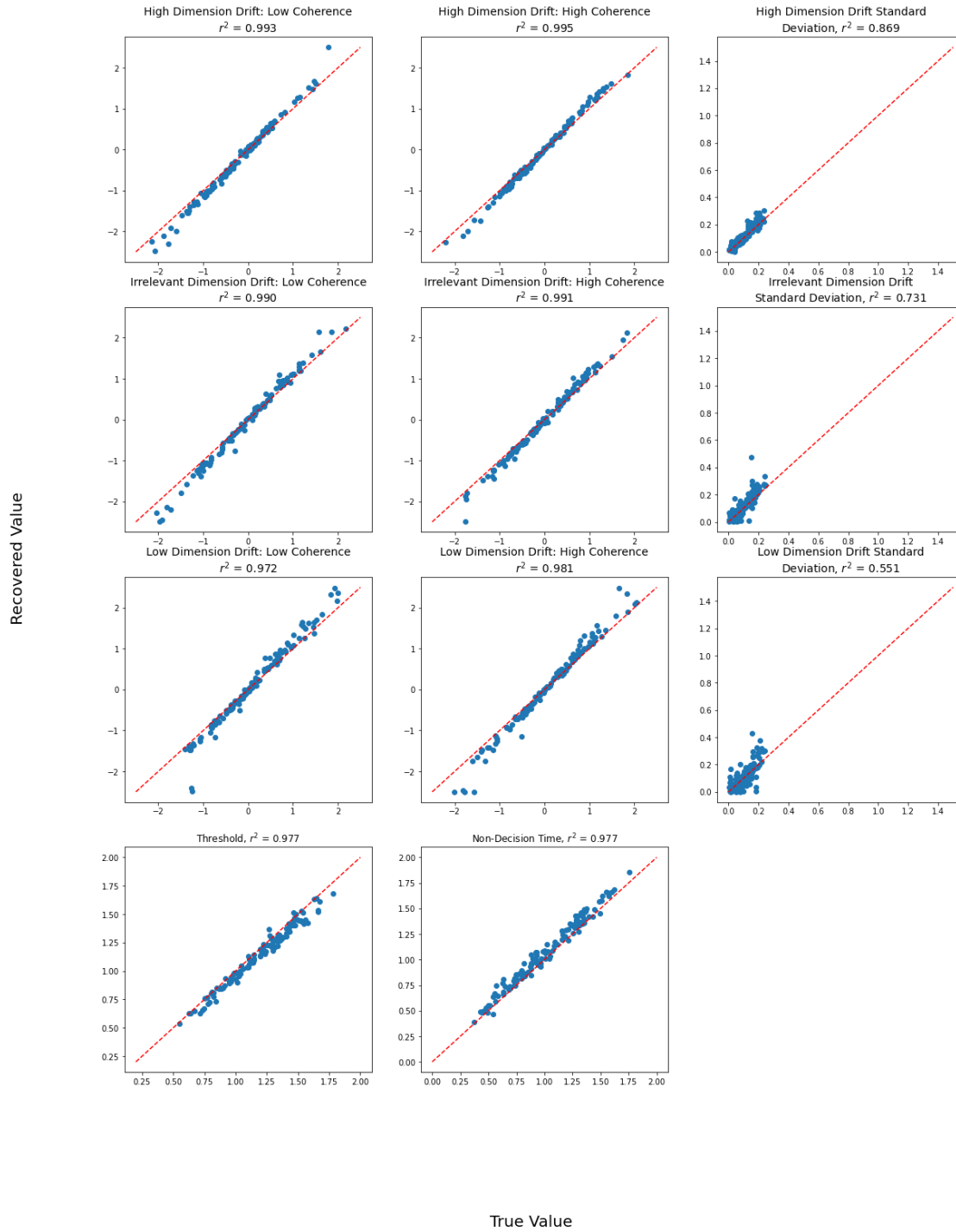


Figure 15: Scatter plots of each parameter's recovery across all 120 synthetic trials for the parallel model. The red line is the line of perfect recovery.

Parameter Recovery and Posterior Predictive:

A total of nine different models were fit to the data. These were versions of the sequential, parallel, and trade-off models that each have one of three boundary types. The three boundary types are constant bounds, angle bounds, and Weibull bounds. The angle boundary type is a linearly decreasing bound while the Weibull boundary type is a Weibull survival function. These models were fit with varying levels of complexity, but the most complex ones were used for analysis as they appeared to provide the best fit as indicated by each models' deviance information criterion and qualitative comparisons. The main ones used for the present analysis vary each of the drift parameters by their respective dimension's task and coherence for each subject, making it a hierarchical model. These were also the most complex models.

While nine models were fit, inference for models with collapsing bounds systematically recovered flat bounds, suggesting that the addition of urgency to the model is not (at least in the form of strictly collapsing bounds) necessary. Interestingly, the “theta” parameter for the angle boundary was recovered at the negative lower bound of -0.1, which would suggest a boundary expansion over time. This has motivated the development of a new flexible boundary that expands and then collapses, though we leave the respective analysis for future work.

All of the candidate models were fit to data with one chain using 5000 Markov Chain Monte Carlo samples, of which 500 such samples were discarded as a burn-in sample. All models produced surprisingly similar posterior predictives (see Figures 16, 17, and 18), which also poses a challenge to favoring one model over another. Two of the clearest failures of all the models are an underestimation of the density of fast reaction times, especially in the “HH” coherence condition (See the reaction time histograms and quantile probability plots in Figures 16, 17, and 18 compared to those in Figure 8). The second failure is having more high dimension

errors than low dimension errors in the “LH” coherence condition. The behavioral data shows low dimension errors occurring more in all conditions. This failure appears to be from an overestimation of high dimension errors. The sequential and parallel models appear to underestimate the low dimension errors, but the trade-off model seems to better capture the low dimension errors and shows a smaller distance between low and high dimension errors (comparing error rate plots of Figure 16 to those of Figure 17 and 18). Another failure of the models is that high dimension errors do not vary by changes in low dimension coherence. This is a counterintuitive artifact found in the behavioral data, which is highlighted earlier. Since this behavior was not expected, the models were not designed to capture it.

Otherwise, the models were quite successful at reproducing the behavioral data. The reaction time quantiles in the coherence conditions and error rates are very well reproduced in all three models, as seen by the quantile probability plots and other graphs found in Figures 16, 17, and 18. They also reproduced the greater effects of coherence in the low dimension compared to the high dimension, also seen in the quantile probability plots.

There are also some interesting peculiarities of the models’ posterior predictives. While the previous analysis expected parallel processing to show reaction times in the “LH” and “HL” conditions to be closer to the reaction times in the “HH” condition than the “LL” condition while sequential processing would show reaction times in “LH” and “HL” conditions to be equidistant from the “HH” and “LL” conditions (see Figures 4 and 5), both the sequential and parallel models demonstrated the behavior expected of sequential processing. Additionally, subject 3 shows the behavior expected of parallel processing, yet the parallel model’s posterior predictive of subject 3 still demonstrates the behavior expected of sequential processing. These results

demonstrate that this prior expectation is not valid given the implementation of the models and, therefore, cannot currently diagnose behavior as parallel or sequential.

Another peculiarity of the models is that the sequential model shows lower reaction times in “HL” than “LH” while the parallel and trade-off models show the opposite. The behavioral data also shows the opposite, though it is not statistically significant. This may indicate that the trade-off and parallel models are slightly better at aligning to the data in this respect, though this may not be meaningful.

The trade-off model has both better and worse qualities. It does not overestimate the number of correct answers as much as the other two models, but it underestimates the number of low dimension errors. It also does a better job of ordering the coherence levels, especially in the low dimension error type.

Trade-off Model Plot Bank

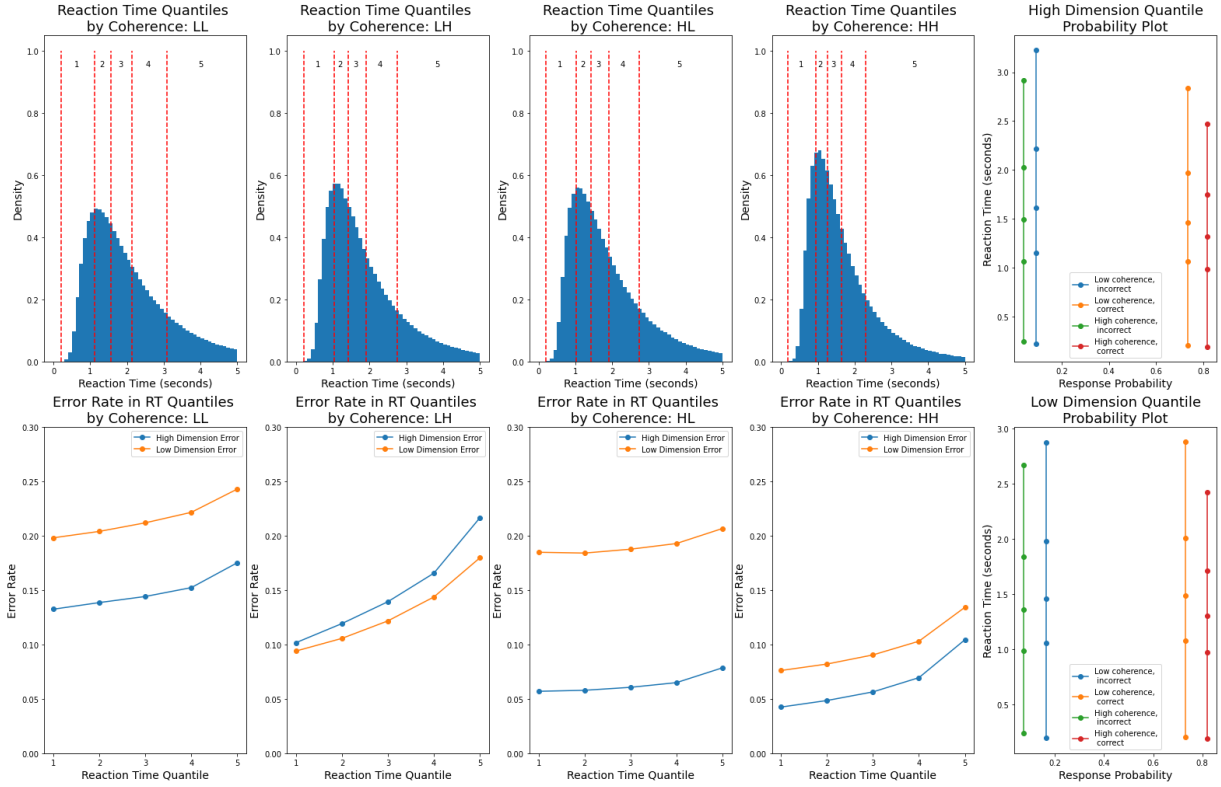


Figure 16: Plot bank of the trade-off model's posterior predictive data. The reaction time histograms and errors by RT quantile are vertically aligned by their coherence condition. High coherence in the high dimension, high coherence in the low dimension (HH); low coherence in the high dimension, high coherence in the low dimension (LH); high coherence in the high dimension, low coherence in the low dimension (HL); low coherence in the high dimension, low coherence in the low dimension (LL). The red lines in the histogram plots mark the beginning of a new quantile, starting at the minimum reaction time (followed by the 0.2, 0.4, 0.6, and 0.8 quantiles). The error rate plots show the proportion of errors in each quantile, calculated by averaging the error rates in a quantile over all the subjects. The two quantile probability plots differ by the high dimension and low dimension. These plots have five dots in each line that represent the same quantiles as seen in the reaction time histograms.

Parallel Model Plot Bank

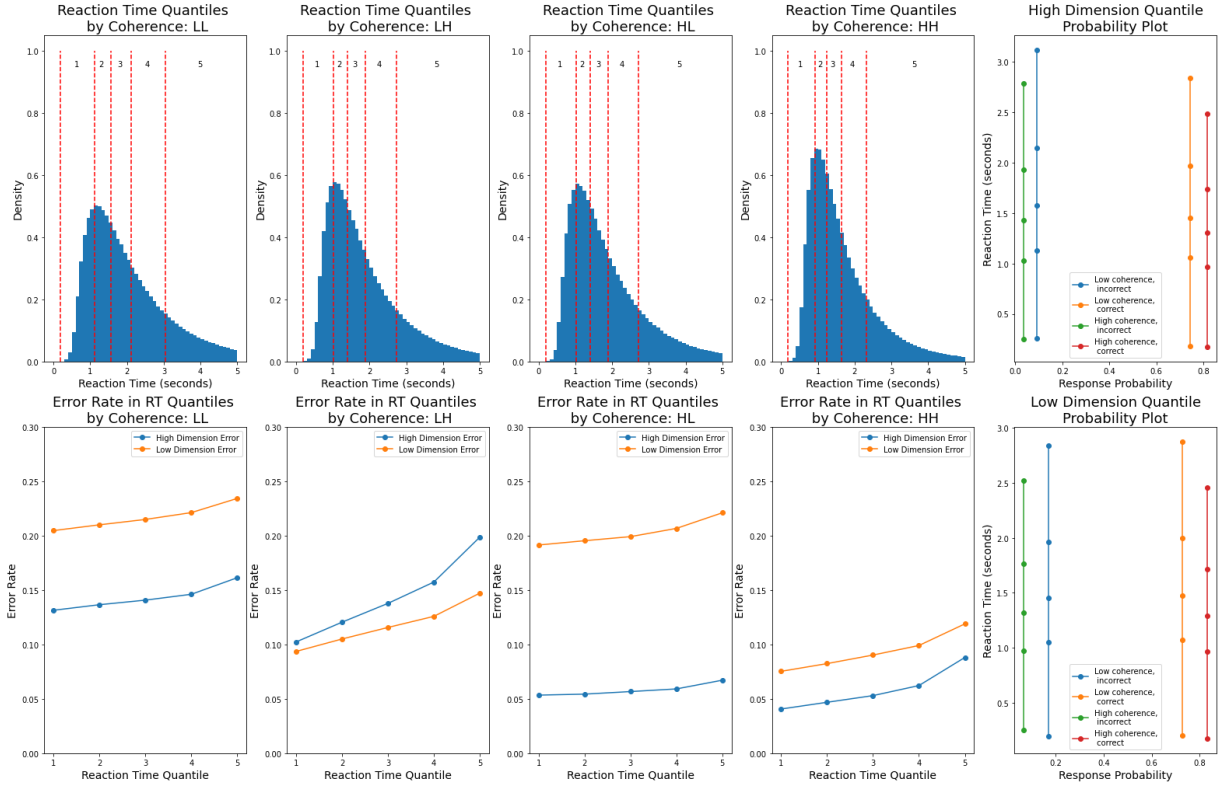


Figure 17: Plot bank of the parallel model's posterior predictive data. The reaction time histograms and errors by RT quantile are vertically aligned by their coherence condition. High coherence in the high dimension, high coherence in the low dimension (HH); low coherence in the high dimension, high coherence in the low dimension (LH); high coherence in the high dimension, low coherence in the low dimension (HL); low coherence in the high dimension, low coherence in the low dimension (LL). The red lines in the histogram plots mark the beginning of a new quantile, starting at the minimum reaction time (followed by the 0.2, 0.4, 0.6, and 0.8 quantiles). The error rate plots show the proportion of errors in each quantile, calculated by averaging the error rates in a quantile over all the subjects. The two quantile probability plots differ by the high dimension and low dimension. These plots have five dots in each line that represent the same quantiles as seen in the reaction time histograms.

Sequential Model Plot Bank

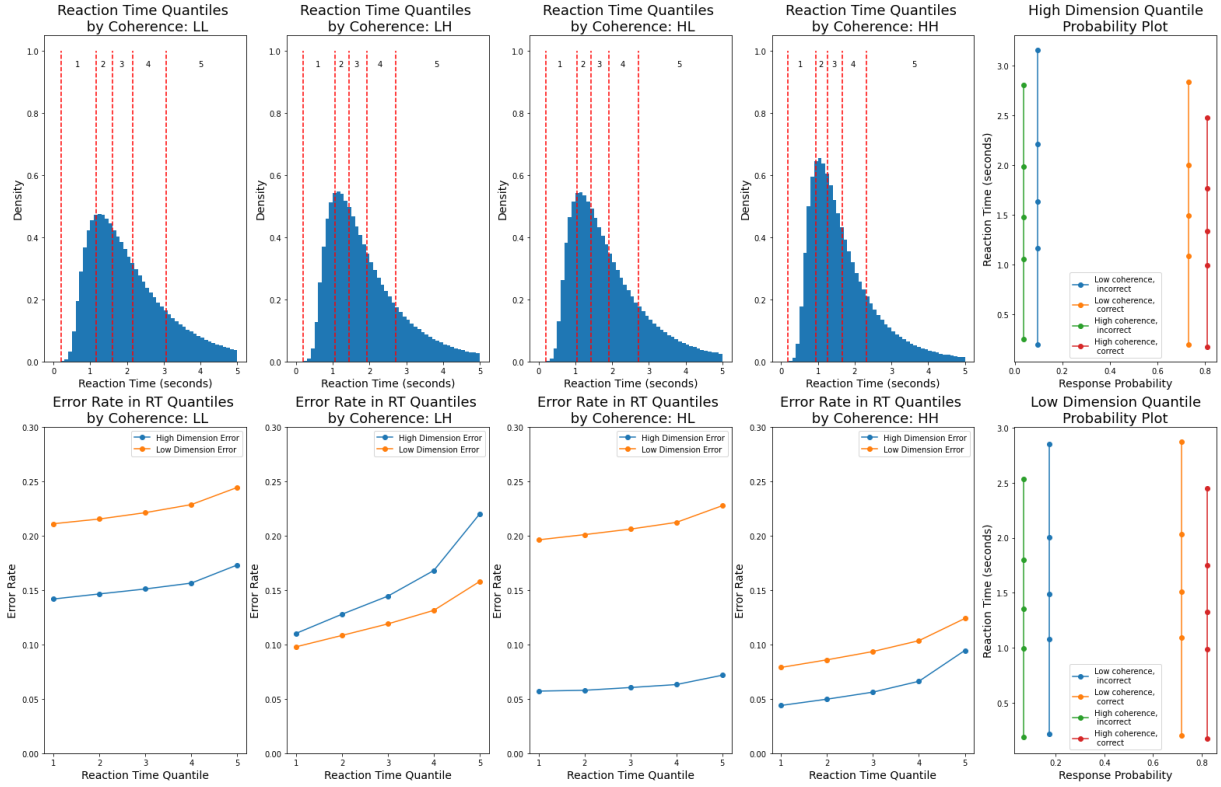


Figure 18: Plot bank of the sequential model's posterior predictive data. The reaction time histograms and errors by RT quantile are vertically aligned by their coherence condition. High coherence in the high dimension, high coherence in the low dimension (HH); low coherence in the high dimension, high coherence in the low dimension (LH); high coherence in the high dimension, low coherence in the low dimension (HL); low coherence in the high dimension, low coherence in the low dimension (LL). The red lines in the histogram plots mark the beginning of a new quantile, starting at the minimum reaction time (followed by the 0.2, 0.4, 0.6, and 0.8 quantiles). The error rate plots show the proportion of errors in each quantile, calculated by averaging the error rates in a quantile over all the subjects. The two quantile probability plots differ by the high dimension and low dimension. These plots have five dots in each line that represent the same quantiles as seen in the reaction time histograms.

Discussion:

All of the models successfully reproduced many of the behaviors seen in the observed data. However, the models share similar weaknesses, and the differences between them are minuscule. This poses a problem for model selection. The only model selection inferences to draw are that the collapsing bounds, angle and Weibull, do not provide substantial benefit over constant bounds, which is somewhat surprising given the previous analysis by Kim et al., (2017) predicted that adding urgency via collapsing bounds to the sequential model would improve the fit by allowing the models to better capture the increase in error rate as reaction time increases. All of the models, however, reproduced the increase in error rate as the reaction time increases, using constant bounds (see error rate plots in Figures 16, 17, and 18).

When choosing how to fit the models, p_{outlier} , a variable that determines how much of the data should be treated as outliers during model fitting, was initially set to 0.05, but all of the models and data reported are from p_{outlier} set to 0. When p_{outlier} was 0.05, the reaction time histograms better captured the high relative number of fast reaction times than the current models, but these models also introduced a strong artifact not seen in the behavioral data. The error rates and mean reaction times in the last quantile of the data were far higher than in the observed behavioral data. This is due to sampling outliers from a uniform distribution with a maximum of 10 seconds. Therefore, 5% of the data came from this uniform distribution and grossly affected the last quantile of data. Perhaps sampling from a different distribution would fix this issue, but more research must be done to ascertain the best option. Due to this artifact, p_{outlier} was set to 0.

Another debatable choice in fitting the models was varying the drift parameters by the task in that dimension. Varying the model by coherence appears very important to the overall fit,

but varying the models by task is less clear. The recovered parameters for each task vary, though not drastically. The observed data indicates some differences in behavior in each task, but the effects are inconsistent and altogether unclear. The models perform better varying by task, but this may be unnecessarily adding complexity to the fits.

Of all the models, the trade-off model appears to be slightly better since the low dimension errors in the “LH” coherence condition better match the observed data. That being said, the model still fails to account for high dimension error rates varying by low dimension coherence. While the models could be set to vary high dimension slope by low dimension coherence, we had no a priori reason to implement such a manipulation. Perhaps revising the models to take into account low dimension coherence for the high dimension drifts would address this issue, but this should be informed by prior theory.

Chapter 4: Conclusions

The sequential, parallel, and trade-off models of evidence accumulation all reproduced the data fairly well. Differences between each model are small, but the trade-off model appears to fit the best. Given the poor recovery of the dampen parameter, the fit of the trade-off model may not be very trustworthy. If the dampen parameter is not being well recovered, then perhaps another aspect of the trade-off model accounts for the better alignment of low dimensional errors in the “LH” coherence condition. The trade-off model’s parameter recovery for the dampen parameter is close to 1, the parameter setting that turns the model maximally towards parallel processing. The trade-off model does not calculate parallel processing in the same way as the parallel model, and perhaps this difference gives it a better fit. The difference is that the trade-off model modulates low dimension evidence accumulation by the particle location of the high dimension while evidence is still being accumulated in the high dimension. To test this possibility, the trade-off model should be refitted with the dampen parameter fixed to 1.

The models demonstrated that flexible boundaries did not improve the fit. While this goes against the expectation of the previous analysis, the models fit separate drifts for the different dimensions, which likely allowed enough flexibility to not require collapsing bounds. Future analysis could explore setting one drift rate for all dimensions, but each dimension always has a different task from the other dimensions, so this experimental design is ill-suited for such an analysis given that there is no control for task difficulty.

The common failure that is not captured in the current models is the greater number of low dimension errors in the “LH” coherence condition compared to the high dimension errors. While varying high dimensional drift by low and high dimension coherences would likely address the issue, such a change should be motivated by a proper hypothesis. The failure is

largely driven by the models overestimating high dimension error. One would suspect that adding collapsing boundaries would give the model the necessary freedom to raise low dimension errors without raising high dimension errors, but the recovered parameters do not suggest that.

A major limitation of this study is the lack of knowledge surrounding the original dataset. While this study was able to recreate the plots and effects seen in the prior analysis of the data, there exists a concern that the low dimension conditions may actually be the irrelevant dimension when the participants make a high dimension error. If this is the case, then the error would also exist in the prior analysis. Future analysis should ensure that everything is labeled and presented consistently. Another limitation is the `p_outlier` sampling. Outlier data is likely not coming from a uniform distribution. Models with `p_outlier` may be able to better capture the desired behavior, so addressing this issue may yield better results.

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